Omolola Adejoke Ogbolumani

Research Interests: Human-in-the-Loop Artificial Intelligence, Food-Energy-Water Nexus Optimization, Machine Learning for Sustainable Systems

1. EDUCATION

The University of Chicago
The Eric and Wendy Schmidt AI in Science Postdoctoral Fellowship, Jan. 2026 – Now
Advisor:
Junhong Chen, Prof. Dr. (junhongchen@uchicago.edu; The University of Chicago-Pritzker School of Molecular Engineering; Argonne National Laboratory)

University of Johannesburg, South Africa
Doctor of Philosophy, Faculty of Engineering and The built Environment, April 2019 – October. 2022
Title of Thesis: A Hybrid Decision-making Framework for Optimal Resource Allocation of Food, Energy and Water Nexus
Advisor: Prof. Nnamdi Nwulu

University of Lagos, Nigeria
Master of Science Electrical and Electronics Engineering, Department of Electrical and Electronics Engineering
Jan. 2006 – Dec. 2006

Lagos State University, Ojo, Lagos, Nigeria
Bachelor of Science, Electronics and Computer Engineering, Faculty of Engineering, Sept. 1995 – Jun. 2002

2. PROFESSIONAL QUALIFICATIONS & MEMBERSHIP OF BODIES

- **COREN** Registered Engineer (R.Engr.), Reg. No: 30,670
- **NSE** Member, Nigerian Society of Engineers (MNSE) Reg. No. 32858
- **NCS** Member, Nigeria Computer Society (MNCS) Reg. No. 09066
- **IEEE** Member, Institute of Electrical and Electronics Engineers Reg. No. 80516145

3. WORK EXPERIENCE

University of Lagos, Akoka, Lagos, Nigeria

- *Lecturer I* – 2022 to 2025
- *Lecturer II* – 2017 to 2022

Yaba College of Technology, Yaba Lagos, Nigeria

- *Lecturer II* – 2015 to 2017

Industry Experience

- **Jobix Computech Consult Ltd.** – Consultant, Project Manager, Computer Engineer 2002 –2015

3. AREAS OF SPECIALIZATION

- Smart Sustainability, Mathematical Optimization, Food, Energy, and Water Nexus, Internet of Things (IoT), Machine Learning and Blockchain Technology

4. CURRENT RESEARCH

My research focuses on developing human-centered AI systems that integrate machine learning and neural networks to optimize complex sustainability challenges, particularly within the food-energy-water nexus. I investigate how human expertise can be effectively incorporated into AI-driven design and optimisation processes to create more robust and practical solutions for resource management. This work is positioned within the context of Industry 5.0, emphasising the collaborative relationship between human intelligence and artificial intelligence in addressing global sustainability challenges.

5. TECHNOLOGY SPECIALTIES:

Optimization Algorithms for Complex Systems
Sustainable Technology Integration
Machine Learning and Neural Network Architectures

6. PUBLICATIONS

- **Ogbolumani, O. A**., & Adekoya, M. **(2025)**. Intelligent Waste Management Optimization Through Machine Learning Analytics. *Journal of Science Research and Reviews*, *2*(1), 7-26. https://doi.org/10.70882/josrar.2025.v2i1.25
- Ajibola A & **Ogbolumani, O (2024)** Smart Battery Storage Integration In An IoT-Based Solar-Powered Waste Management System. *Journal of Digital Food, Energy & Water Systems, 5*(2). **https://doi.org/10.36615/hrewq861**
- **Ogbolumani, O. A**., & Nwulu, N. I. **(2024)**. Environmental impact assessment for a meta-model-based food-energy-water-nexus system *Energy Reports*, 11, 218-232
- **Ogbolumani, O. A**., & Godfrey, D. **(2024).** Nexus OF Wireframe, 3D Model, AND Simulation For The Development Of An Intelligent Waste Management System . *FUDMA Journal of Sciences*, *8*(6), 328 - 342. https://doi.org/10.33003/fjs-2024-0806-3020
- Ajibola A & **Ogbolumani, O (2024)** Smart Battery Storage Integration In An IoT-Based Solar-Powered Waste Management System. *Journal of Digital Food, Energy & Water Systems, 5*(2). **https://doi.org/10.36615/hrewq861**
- **Ogbolumani, O.,** & Mabaso, B. **(2023).** An IoT-Based Hydroponic Monitoring and Control System for Sustainable Food Production. *Journal of Digital Food, Energy & Water Systems*, *4*(2).
- Adeleke, I. A., Nwulu, N. I., & **Ogbolumani, O. A. (2023**). A Hybrid Machine Learning and Embedded IoT-based Water Quality Monitoring System. *Internet of Things*, 100774
- Olorunfemi, B. O., Nwulu, N. I., & **Ogbolumani, O. A. (2022)**. Solar panel surface dirt detection and removal based on arduino color recognition. *MethodsX*, 101967.
- Olorunfemi, B. O., **Ogbolumani, O. A.**, & Nwulu, N. **(2022)**. Solar Panels Dirt Monitoring and Cleaning for Performance Improvement: A Systematic Review on Smart Systems. *Sustainability***,** 14(17), 10920.
- **Ogbolumani, O. A**., & Nwulu, N. I. (**2022**). A food-energy-water nexus meta-model for food and energy security. *Sustainable Production and Consumption*, *30*, 438-453.
- **Ogbolumani, O. A.,** & Nwulu, N. I. (**2021).** Multi-objective optimisation of constrained food-energy-water-nexus systems for sustainable resource allocation. Sustainable Energy Technologies and Assessments, 44, 100967.
- **Ogbolumani, O. A**., & Nwulu, N. **(2020)**. Integrated Appliance Scheduling and Optimal Sizing of an Autonomous Hybrid Renewable Energy System for Agricultural Food Production. In Advances in Manufacturing Engineering (pp. 651-660). Springer, Singapore.

7. PATENTS

- An Intelligent Real-Time Monitoring Waste Management System - NG/P/2018/13
- An Advanced IoT Driven On-Demand Waste Tracking, Recycling and Disposal System – F/PT/NC/2024/13207

8. REFEREES

Prof. Nnamdi Nwulu

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Johannesburg, South Africa.

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Prof. Chika Yinka-Banjo

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